leveraging on-demand 3D printing and blockchain technology.

According to a press release from Air New Zealand Communications, their aircrew on a Boeing 777-300 aircraft noticed that a business premier bumper part—that sits between a seat and monitor to prevent the screen from damaging the seat when it's pushed back to default position—needed replacement and ordered a digital aircraft part file from ST. The file was sent to Moog's printer and the 3D printed part could be installed within a few hours ahead of the flight's scheduled departure. The traditional process would have taken days, which would have resulted in a huge revenue loss.

In another example, ERIKS, a company that offers technical components and related services to various industries internationally, began using Ultimaker S5 3D printers to customise safety parts, iterate faster, and scale up its production. The company now 3D prints seals, gaskets, valves, safety tools, prototypes as well as end-parts that are hard to create, ensuring not only quality but also reproducibility and reliability. Besides saving over € 350,000 through such initiatives, the company could successfully eliminate safety hazards in their facilities.

Customisation and personalisation have also played an important role in providing 3D printing an edge over traditional manufacturing. This can be seen from Wiivv Technology's concept of custom-fit 3D printed insoles for their footwear. After users purchase sandals of their choice, they are given a special link to take foot measurements using the app and a white paper. Each foot of an individual is digitally mapped with over 200 points to create a unique file of an insole for 3D printing. The custom-fit product is then sent to the customer.

In India too, there have been numerous instances where 3D printing was chosen as the ultimate solution for a problem. For instance, for the first time in the country last year, Bengaluru-based Yellow Ribbon team collaborated with Osteo3D and implantcast GmbH in Germany to treat a patient suffering from a rare bone tumour known as Chondromyxoid fibroma (CMF) of pelvis that's typically found in children and young people. After removing the tumor completely, the team used CT and MRI scans of the patient to create a digital model for surgical planning and designing a custom 3D printed pelvic implant. The operation was successful and prevented the patient from undergoing an amputation.

With the price of printers falling down as well as new emerging technologies and applications, the 3D printing business has a bright future for industries worldwide. It has the potential to change how businesses operate and cater to the needs of their customers. As of now, in industries where not all parts can benefit from embracing the 3D printing process, a hybrid approach is suitable. Even beyond businesses, 3D printing is slowly making its way into areas like homes and emergency sites. EFY

*—Ayushee Sharma*